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Fulfilment of last-mile urban logistics for sustainable and inclusive smart cities: a case study conducted in Portugal

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ABSTRACT

While industry tends towards mass personalisation and instant deliveries, the last mile of urban logistics is being challenged to decrease the number of vehicles in circulation and the distances they travel in city centres. The COVID-19 pandemic helped expose the inefficiency of cities in fulfilling citizens' real-time needs. Moreover, the first aim of this paper is to understand the barriers which policymakers face in providing a personalised response to citizens' needs and the second to ascertain how they can proactively serve their communities. In line with these concerns, empirical evidence was collected through a questionnaire to Portuguese policymakers, and the results were discussed in a focus group with experts. The results suggest a lack of tools to enable real-time visualisation and study of scenarios for implementing and organising means of delivery and storage. Moreover, although policymakers feel confident in their capacity to manage the last mile, they would struggle to operate autonomously. Therefore, this paper conceptualises an initial algorithm based on the dynamic collaboration of stakeholders and sharing of resources to guarantee fulfilment of citizens' necessities. Furthermore, future discussions shall emerge about the relationships and technical standards between stakeholders to provide the necessary logistical means for city management and operations.

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Real-time fulfilment; instant deliveries; stakeholders collaboration; resources sharing; city management

1. Introduction

Urban logistics or city logistics refers to the movement of urban goods resulting from logistics decisions that are intrinsically related to the existing demand and the behaviour of economic agents (Dablanc 2007). The concept is usually described as optimising transport activities in urban areas (Taniguchi et al. 2001). Furthermore, it is linked to the proper planning of the distribution of goods within a city, and managing and controlling freight movements, considering integration and coordination among stakeholders (Amaral and Aghezzaf 2015; Morfoulaki et al. 2015).

Over the years, several reflections have emerged on how urban freight activities could work toward cities' sustainability while increasing inclusion. For example, Brusselsaers, Mommens, and

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Macharis (2021) present a participatory stakeholder framework that facilitates and supports logistics to improve mobility and reduce the negative impact of construction sites on the surrounding community. Usually, these reflections were attached to the introduction of policies by government bodies to force companies to change their actions (Anderson, Allen, and Browne 2005; Muñuzuri et al. 2005). In addition, multiple studies envision participatory decision-making processes to support urban freight transport policies (Le Pira et al. 2017; Marcucci et al. 2017). Thus, sustainability and inclusion are concerns and trends in this area since there is the ambition to include citizens and the rest of stakeholders in decision-making processes, while using their resources and availability to fulfil personalised requests (Correia, Teixeira, and Marques 2020, 2021a).

Urban logistics tends to focus on the last mile mostly because of phenomena of urbanisation and e-commerce. While consumer demand pushes for personalisation, entities' delivery and storage resources put pressure on the last step of the supply chain. Local authorities are growing concerned about the concentration of parcel deliveries (Ducret 2014). Cities are challenged by the engagement and integration of stakeholders to combat the spread of companies' vehicles and warehouses in the city centre (Lagorio, Pinto, and Golini 2016). This is because unintegrated logistics activities challenge cities' sustainability (Bibri and Krogstie 2017). Furthermore, it represents between 8% and 18% of urban traffic and 21% of CO² emissions in residential areas (Nocerino et al. 2016; Russo, Rindone, and Panuccio 2016). Moreover, there is a need to reduce unadded value transportation activities and decrease the number of vehicles in circulation by combating the inefficiency of mobility and logistics means of transportation (Boysen, Fedtke, and Schwerdfeger 2020; Le Pira et al. 2021). Different solutions can be considered across the different phases of the supply chain. Several authors are studying ways to decrease the impact, namely Wang et al. (2020), who propose introducing eco-packages, or De Oliveira, Albergaria, and Bandeira (2017) who through a systematic literature review, note a trend for the implementation of smaller and lighter vehicles for last-mile deliveries in urban areas (namely bicycles and tricycles or light commercial vehicles).

Moreover, to provide a real-time answer, there is a need for information on existing resources, such as delivery and storage assets. On top of that, vehicle routing and location selection of warehouses are important logistics decisions to allow a faster response while decreasing the number of resources needed. Thus, novel approaches are necessary to meet consumer demand while improving the organisation of current resources. Individual and proprietary supply chains may be replaced by dynamic models where policymakers can play a crucial role in establishing and regulating the relationships and communication between stakeholders.

Policymakers have already made use of the emergence of smart cities to fill the existing lack of urban data on which to base their decisions (Batty et al. 2012; Chourabi et al. 2012; Hall et al. 2000; Harrison and Donnelly 2017). The internet of things (IoT) has increased authorities' responsiveness and promoted the creation of standards (for the integration of different platforms and applications) to allow cities to have a centralised operating scenario (Jin et al. 2014; Mulligan and Olsson 2013). This has permitted cities to create control centres (also known as urban platforms) to allow policymakers to base their decisions on real-time analytics and predictive models associated with historical information (Correia, Teixeira, and Marques 2021c, 2021b; Gutiérrez et al. 2018; Kitchin 2014; Townsend 2000). Nevertheless, there is a lack of last-mile urban logistics considerations to decrease the number of delivery vehicles in city centres. The integration of these concerns into the control centres, and the guarantee of interoperability which is capable of creating an omnichannel where collaboration is dynamic and proactive are vital aspects.

Several studies on this issue can be found in the literature about existing challenges, namely the difficulty of accessing urban data and the integration of different sources for multimodal approaches (Perboli et al. 2018). Thus, measures and policies performed by decision-makers are needed for sustainable and efficient management of supply chain activities in urban areas (Paddeu et al. 2018). These policies must consider solutions with positive impacts on urban mobility while contributing to new business models (Cassiano and Bertoncini 2021). These approaches are

inspired by the concept of Logistics 4.0 which is derived from the Industry 4.0 as its background. The concept was characterised by Timm and Lorig (2015) as pivoting from being hardware oriented to software oriented. Thus, it represents the fulfilment of individualised customer demands, supported by intelligent systems to achieve a significant automation and data visualisation (Facchini et al. 2020). Nevertheless, there is still a lack of application in integrating and allocating resources to meet the population's needs (Wang et al. 2021).

Therefore, this study seeks to answer the questions 'Are cities capable of meeting citizens' real-time needs?' and 'How can they do that while reducing the number of vehicles in the last mile?'.

To answer these research questions, a two-folded objective empirical study was conducted within the Portuguese context. On the one hand, the objective was to understand the readiness of policymakers and the challenges they face in providing a real-time response to citizens' needs. On the other hand, the goal was to conceptualise an initial algorithm to support collaborations on the last mile while decreasing the number of vehicles in city centres. As result, a model has been designed which assumes the dynamic cooperation of stakeholders and their resources to proactively position them according to the demand forecast to meet existing real-time events. The aim of this model is to correctly allocate stock to potential storage assets depending on the needs and characteristics of the population. This will allow faster deliveries to be provided and the existing isolation gap to be filled.

The paper is structured as follows: in section 2, a literature review is performed in order to have background information on last-mile deliveries, the state of the art of vehicle routing, and the selection-location problem. In section 3, a two-step empirical research that considers a questionnaire to policymakers and a focus group with experts is detailed. In section 4, on the one hand the results of the quantitative and qualitative analyses are presented, while the significance of the characteristics of respondents (gender, age, city dimension and their position) are evaluated. On the other hand, a four-step solution is designed based on the empirical findings. The city of Lisbon is used as case study to demonstrate the various steps from the analysis of historical citizens' data to the spatial indexing and clustering to find the best candidate points for storage and delivery. Section 5 presents the discussion of the results. Finally, section 6 highlights the conclusion, limitations, and future work.

2. Theoretical background

Last mile is considered in academia as the least environmentally friendly and efficient stage of the supply chain, comprising 28% of the total delivery cost (Gevaers, Van de Voorde, and Vanelslander 2011; Wang et al. 2016). Ranieri et al. (2018) reviewed the literature and indicated collaborative urban logistics and optimisation of transport management and routing as the main innovations to reduce transport costs and inefficiency. Thus, these are expected to be flexible and multi-modal (Prause and Atari 2017). Moreover, co-modality (or crowd shipping) can offer limitless combinations of transport modes that can be fully dedicated to the transportation of goods or allocated simultaneously to other tasks due to their availability or underutilised capacity (Eiichi and Yasushi 2004; Gatta et al. 2018; Kin et al. 2018). A sharing economy, open cross-company network based on win-win last-minute business collaboration schemes is being increasingly considered to optimise urban transport (Ducret 2014; Kirch, Poenicke, and Richter 2017; Nathanail, Gogas, and Adamos 2016; Oztemel and Gursev 2020) and different approaches are being considered. For example, Ghaderi et al. (2022) propose a framework that considers people daily travels, in contrast to the dominant crowdshipping model in which individuals perform dedicated delivery trips. Thus, new studies are emerging to find the best localizations for urban micro centres for last-mile delivery using smooth moods of transportation based on real-time data and city characteristics (Rudolph et al. 2022). Nevertheless, these are still early-stage, high-level reflections and proposals with little application. In addition, the application of this understanding of how policymakers will manage last-mile fulfilment is still lacking.

The inherent challenge of logistics fulfilment in the research questions demands a review of the subjects related to the delivery and storage of goods. Therefore, this section aims to provide readers with a glimpse of existing literature about urban distribution focused on the last mile. Thus, first an overview is given of the evolution of last-mile delivery models and optimisation rules. Second, selection and location problem methods are reviewed, focusing on the placement of warehouses and storage facilities.

2.1. Last-mile fulfilment and instant deliveries

The classical travelling salesman problem (Jünger, Reinelt, and Rinaldi 1995), in which the vehicle starts from a warehouse and visits several customer locations to minimise the total distance travelled, evolved to the single depot vehicle routing problem (SDVRP). The goal was to reduce the distance which vehicles travelled while meeting customer demand and operating constraints (Dantzig and Ramser 1959). Later, the multi-depot vehicle routing problem (MDVRP) extended the SDVRP model, or VRP, by dispersing multiple depots from which multiple vehicles can originate (Gillett and Johnson 1976). After this emerged a variant of the MDVRP called the min-max multi-depot vehicle routing problem (min-max MDVRP). This model aimed to minimise the total distance travelled by each vehicle instead of the entire distance, promoting a more equitable load-sharing (Carlsson et al. 2009; Venkata Narasimha et al. 2013). Recently, Wang et al. (2022, 2021) proposed the collaborative multicentre vehicle routing problem with time windows and mixed deliveries and pick-ups (CMVRPTWMDP).

The evolution of the vehicle routing problem is intrinsically connected to market demand and the tendency towards providing instant deliveries. This refers to on-demand delivery within two hours by independent contractors or couriers via a digital platform (Dablanc et al. 2017). The high time-sensitivity concept, immediately executed after orders are placed, is developing rapidly (Gu et al. 2020). First- and last-mile optimisation (referring to the end or beginning of the supply chain) has taken on greater importance due to the wish of people to have instant deliveries (Bányai, Illés, and Bányai 2018). This poses a significant challenge to the traditional supply chain and existing resources to meet the schedule. Moreover, the sharing economy (dynamic and short-term collaborations with subcontractors, where their earnings are attached to a direct percentage of each service provided) has set the foundations of emerging innovative models (Li et al. 2020).

The fifth wave of logistics, also called consumer logistics, encompasses three key elements: omnichannel retailing, the internet of things, and 3D printing (Rimmer and Kam 2018). The supply chain will move towards local realisation and the real-time urban adjustment based on the study of the most appropriate locations to store goods to deliver the goods to citizens (Shi, Liu, and Zhang 2021; Srivatsa Srinivas and Marathe 2021). Correia, Teixeira, and Marques (2021a) detail this new paradigm on the last mile that can bring manufacturing closer to consumers (using 3D printing technologies) to allow the real-time fulfilment of personalised requests. Nevertheless, until there is a network of manufacturers to produce the product in real time, the supply chain needs to be arranged to ensure the goods are available in the last mile. Thus, to achieve this paradigm, new logistics models based on transport co-modality and sharing of mobile stock points must emerge (Daugherty, Bolumole, and Grawe 2019; Lim, Jin, and Srai 2018; Srivatsa Srinivas and Marathe 2021; Taniguchi, Thompson, and Yamada 2016).

Furthermore, two main options to bridge the 'last mile' are indicated: pick-up points and home delivery (Daduna and Lenz 2005; Morganti, Dablanc, and Fortin 2014). A hybrid or integrated model may emerge where similar pick-up points can work as decentralised warehouses. Machado et al. (2021) propose public transportation (bus network) to act as an integrated urban logistics option combining cargo flow with passenger transportation. Srivatsa Srinivas and Marathe (2021) propose a mobile warehouse based on a moving truck with stock inventory. In addition,

autonomous vehicles are expected to disrupt passenger transportation and delivery fulfilment, combating population ageing and isolation and promoting inclusion of the disabled (Hwang et al. 2021; Prattley et al. 2020; Yang, Yeung, and Feng 2018).

The routing optimisation criteria depend on the assumptions initially considered. Thus, it is necessary to understand how the delivery will be performed to focus on routing optimisation. At the same time, choosing the vehicles to complete the deliveries is one of the most critical decisions at the basis of a logistics model (Crainic, Ricciardi, and Storchi 2004). Sharifi and Khavarian-Garm-sir (2020) reflect on the central issues revealed by the COVID-19 pandemic and the post-COVID recommendations. Among them is ‘greening the transportation and industry sectors’. Villa and Monzón (2021) notice that, in Madrid, during the pandemic period, CO2 emissions related to e-commerce last mile increased by 43.1%. Thus, there is a need for public-private collaboration for transport and logistics operators throughout the supply chain, especially in the last mile, using environmentally friendly vehicles (Settey et al. 2021).

Ranieri et al. (2018), review recent scientific literature contributions on innovative strategies for last-mile logistics and indicate five categories: innovative vehicles, proximity station, collaborative and cooperative logistics, optimisation of transport management and routing, and innovations in public policies and infrastructures.

2.2. Warehouse selection location and spatial indexing

In the early days, Crainic, Ricciardi, and Storchi (2004) found that satellite warehouses reduced trucks’ travel distance in the urban centre but increased the number of vehicles and the total kilometres travelled. Over the years, warehouses moved from the city centres to metropolitan areas. This happened because of land costs and availability, lower tax, and a well-served infrastructure with highway intersections, meaning more considerable distances and an increasing number of vehicles (Dablanc 2014; Guerin et al. 2021). Moreover, land usage distribution regulation and planning influenced logistics sprawl and urban freight transport (Combes 2019).

Multi-criteria decision-making is most widely used to solve the selection location problem (de Carvalho et al. 2020; Özcan, Elebi, and Esnaf 2011). There are many factors to consider according to the structure of the decision problem and the implicit preferences. Nevertheless, it has historically been focused on the cost or service levels. Nowadays, with the increasing emphasis on social responsibility and environmental aspects, sustainability-related criteria have been introduced (He et al. 2017).

Different reflections are spread throughout the literature about the optimal location of urban distribution centres to enhance cities’ sustainability and reduce the number of vehicles in circulation. Some authors observe that these should move away from the city centre (Dablanc and Rakotonarivo 2010; Hesse 2002). Others found more significant environmental savings in moving them closer to consumers (Filippi et al. 2010). Recently, Wygonik and Goodchild (2018) highlighted the need to understand operational details and include them in modelling the use case. As industry strives towards instant deliveries, it is necessary to store goods closer to customers, which may increase the complexity of the supply chain and decrease the size of intermediary warehouses and logistics centres (Correia, Teixeira, and Marques 2021a).

Spatial indexing libraries are scarce topics in the literature. Nevertheless, in 2017 Google made the S2 available, an open-source library that gets nearby objects through spatial indexing (Google 2021b; Pandey et al. 2021). The projection is made by subdividing the planet Earth into a hierarchical decomposition of its entire surface through three-dimensional spherical projections, obtaining and analysing regions according to the desired granularity. Uber also launched its own open-source space indexing library, the H3, which like S2, hierarchically subdivides the Earth’s geographic surface using hexagons and pentagons (Uber 2021). It is based on the projection of the Dymaxion map or Fuller map, which transforms the Earth’s spherical surface into an icosahedron (Atlas of Places 2018; Gray 1994).

These flexible models allow spatial indexing to find the best option to satisfy real-time, constantly changing needs compared to the traditional selection location problem.

This literature review acknowledged that routing optimisation has evolved to reduce the number of resources applied to operations and decrease greenhouse gas emissions. The choice of the type of transportation is one of the crucial takeaways. The paradigm of instant deliveries demands an omnichannel with the capacity to seek the best collaboration available in terms of delivery and storage assets to fulfil a specific request based on co-modality and collaboration schemes. The decentralisation of distribution centres to smaller indexed options improves the adaptative capacity and flexibility of the system. However, it lacks the transference of knowledge to municipalities and decision-makers. Policymakers need preparation and awareness of the logistics demand (Dias et al. 2018). Thus, there is a need to assist them with the understanding of how to setup a dynamic collaboration between stakeholders and citizens in the last mile, and study what the policymakers' role and capacity is to sustainably respond to real-time events.

3. Methodology

Based on the premises of the literature review, the overall goal of the methodology was to understand the challenges that policymakers encounter when meeting citizens' needs, and discuss how they could proactively address the demand. Thus, the methodology followed in this research, sketched out in Figure 1, was divided into two methods: (i) a questionnaire to policymakers and (ii) a focus group with industry experts. Each method addressed a specific research question.

In the first step, the contacts of Portuguese policymakers were collected and addressed via email. Quantitative analysis was performed on the closed questions, while qualitative analysis was performed on the open questions to have their opinions on the existing challenges. In the second step, a focus group was held to discuss the foundations of an algorithm that could assist policymakers' decisions and mark the first step of an integrated urban response. Qualitative analysis was done on the transcript content.

The Portuguese context was chosen based on the authors' knowledge of the sector, the capacity to get the responses of the most significant number of policymakers to the survey and the authors' close relationship with industry experts.

3.1. Questionnaire to policymakers

The aim of the questionnaire was to get policymakers' views in order to understand whether cities can respond to real-time events and if they have tools to move the necessary resources based on predictive analytics.

The questionnaire applied (Appendix) was composed of closed and open questions. It was created using Google Forms and sent individually via email between 6 and 25 September 2021. Structurally, the questionnaire was divided into three parts. First, initial questions aimed to characterise respondents according to their age and gender (other attributes such as the size of their cities and their position were also associated in the third part). In the second part, the goal was to ask policymakers if their cities would be capable of meeting the needs of citizens and how they would depend on private entities. The third part aimed to understand whether municipalities have detailed information about citizens and technological tools that help them organise the urban logistics based on these data. In addition, they were asked if they would value the existence of these tools. Ultimately, the intention was also to understand how they organise urban logistics. On the one hand, quantitative analysis was performed using SPSS software. On the other hand, a thematic analysis of the answers to the open questions was performed using NVivo software.

The questionnaire collected the opinions of Portuguese policymakers. A unique country approach was chosen in order not to have biased results because of different contexts and circumstances among respondents. Therefore, every public contact of Portuguese policymakers of the 308

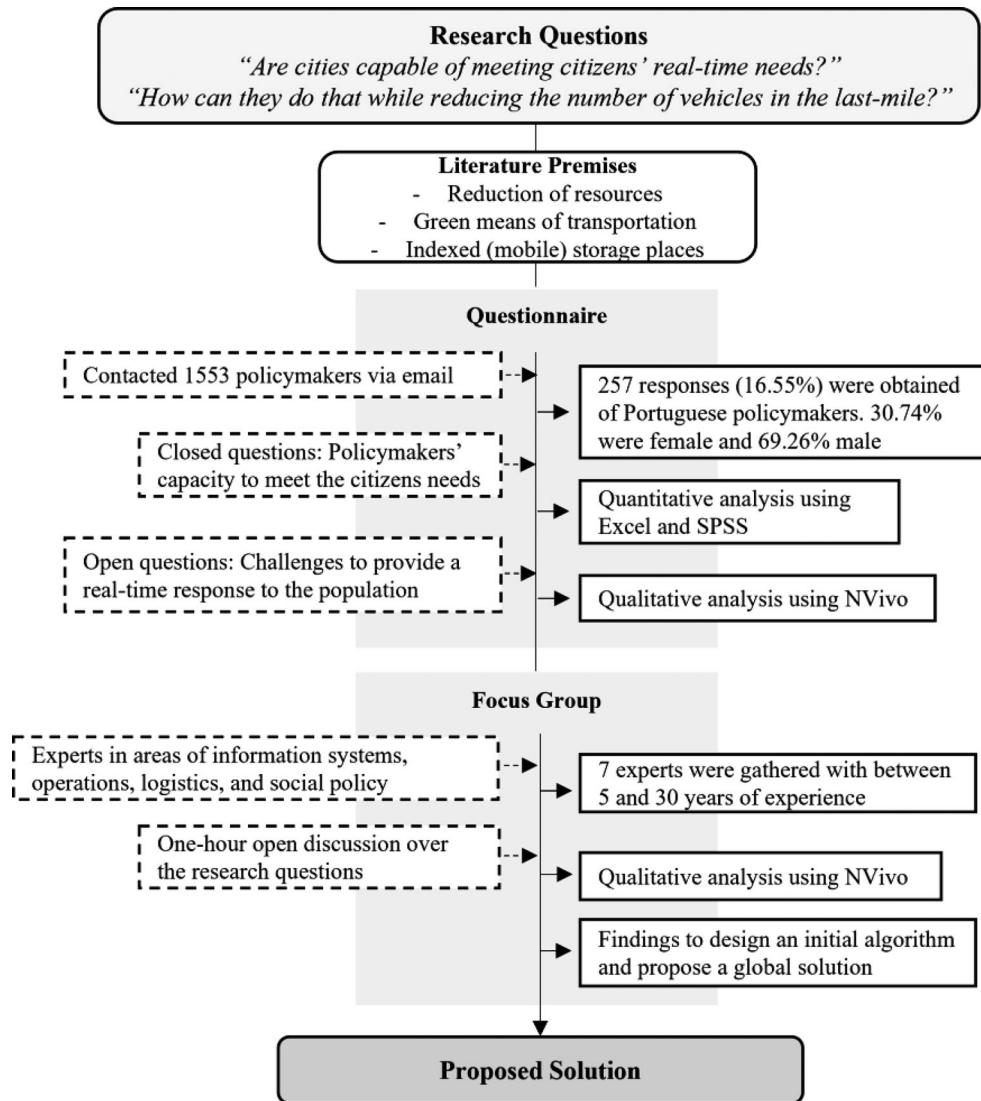


Figure 1. Methodology framework.

municipalities was collected, totalling a population of 1553 contacts. The final sample was composed of 295 responses (19.00%), of which 30.85% were female and 69.15% male. In addition, 46.10% of respondents were aged between 40 and 49, 36.61% aged 50–64, 9.15% aged 25–39, 7.8% over 65 and 0.34% aged 18–25.

As for the respondents’ roles, 26 are mayors, 14 are vice-mayors, 170 are councillors, 36 are heads of department, 19 are assistants or advisors, and 30 are technicians. Moreover, 134 policymakers of cities are represented with less than 25,000 inhabitants, 59 of 25,000–50,000 residents, 47 of 50,000–100,000 inhabitants, 42 of 100,000–200,000 inhabitants, and 13 of more than 200,000 inhabitants.

3.2. Focus group with experts

A focus group was held in which discussion between experts was promoted to propose a solution to answer the research questions. Moreover, the aim was to find the foundations of a tool that could

meet citizens' real-time needs. Therefore, based on the questionnaire findings (about the challenges that policymakers stressed in responding to the population in real time), the experts were asked to think of features that the solution should consider.

The focus group followed the approach put forward by Morgan (1998) and Stewart, Shamdasani, and Rook (2007), characterised by promoting an open and flexible discussion with a collective understanding uncovered by individual interviews, allowing the researcher's direct interaction with the experts. In addition, discipline was exercised in meeting the time restrictions the assertive moderation for the interventions' objectivity. The online focus group lasted one hour, was held via Zoom, and the authors acted as moderators and supervisors. The conversation was recorded, and the content was later transcribed and analysed.

The focus group brought together a heterogeneous group of experts on social policy, information systems, manufacturing, operations management, and logistics. Furthermore, seven experts were gathered with a range five to 30 years of experience, combining the industry vision with the academic and regulation perspective. Table 1 describes the experts.

4. Results

This section, on the one hand, summarises the quantitative analysis' results, provides a qualitative analysis of the questionnaire' open questions, and presents the results from the focus group discussion. Furthermore, it details the model that resulted from the findings of this empirical research.

4.1. Policymakers' capacity to organise urban logistics in the last mile

Table 2 details the results of the quantitative analysis of the central questions of the questionnaire (Appendix). The first aim was to understand if cities would be capable of organising last-mile urban logistics to meet citizens' needs, what their dependence on private entities would be and their need for a logistics support tool. The median was considered as the central tendency measure. Moreover, MDA (median absolute deviation) was considered for the dispersion measure of the variability of the univariate sample of quantitative data. Furthermore, its value was 1 for all questions.

Analysis of Table 2 provides valuable insights that help identify the existing lack of tools to support last-mile urban logistics decisions.

Moreover, when policymakers were asked if cities would be able to organise the last mile of urban logistics, 87.8% of respondents answered 'Yes, totally' or 'More or less' regarding the ability of cities to meet citizens' needs.

Furthermore, the median was 3 on the five-point Likert scale to the question about the dependency of cities on the private sector. Only 6.10% people answered, 'Not at all dependent', which presupposes cities would struggle to operate autonomously in respondents' opinion.

In addition, 76.61% of respondents reported not having tools that enable real-time visualisation and study of scenarios for placing provisional means (delivery and storage). The median answer about the importance of that tool was 4 on the five-point Likert scale. Only 0.88% of respondents answered, 'Not at all important'.

Table 1. Focus group' elements identification.

Expert	Role	Area	Experience
1	European commissioner assistant	Social policy	5 years
2	Professor	Information systems	12 years
3	CEO	Industry 4.0	15 years
4	CTO	Logistics	8 years
5	Software engineer	Future mobility	10 years
6	Professor	Operations management	30 years
7	CEO	Manufacturing and 3D printing	20 years

Table 2. Policymakers' answers about their capacity to organise last-mile urban logistics.

Topic	Question	N	Answers	Median
1. Logistics organisation (Appendix 1 – question 3)	'In your opinion, do you think that cities would be able to organize last-mile urban logistics to meet citizens' needs?'	137	2.71% – '1 No'	3
		122	9.49% – '2 Not much'	
		28	41.36% – '3 More or less'	
		8	46.44 – '4 Yes, totally'	
2. Dependency on private entities (Appendix 1 – question 4)	'How dependent would you be on the responsiveness of the private sector?'	18	6.10% – '1 Not at all dependent'	3
		28	9.49% – '2'	
		126	42.71% – '3'	
		95	32.20% – '4'	
		28	9.49% – '5 Very dependent'	
3. Response capacity (Appendix 1 – question 10)	'Considering citizens' needs had to be met in real time (15 min), how difficult would it be for the city to move the necessary means?'	7	2.37% – '1 Not at all difficult'	4
		28	9.49% – '2'	
		84	28.47% – '3'	
		89	30.17% – '4'	
		87	29.49% – '5 Very difficult'	
4. Tools (Appendix 1 – question 7)	'Do municipalities have technological tools that allow real-time visualization and the study of scenarios to place provisional means (delivery and storage) for the supply of goods?'	69	23.39% – 'Yes'	*
		226	76.61% – 'No'	
5. Importance (Appendix 1 – question 8)	'... how important do you think the existence of a tool with these features would be?'	2	0.88% – '1 Not at all important'	4
		6	2.65% – '2'	
		36	15.93% – '3'	
		91	40.27% – '4'	
		91	40.27% – '5 Very important'	

*It is a nominal variable.

In terms of the level of significance of the variables gender, age, city dimension, and respondents' position, [Table 3](#) summarises the findings when applying the chi-square statistic test to the sample.

The results demonstrate that cities with fewer inhabitants are also less confident about their willingness to organise the last mile. This may be explained by the fact that most Portuguese cities of this size are in rural areas, which presents challenges to operations and their capacity to cover the entire region.

Mayors and vice-mayors are confident about their capacity to move the necessary means. However, they are cautious about meeting citizens' needs in 15 min. Assistants/advisors and technicians are less optimistic, which could mean that cities may struggle to organise their assets.

Table 3. Significant variables' findings.

Question	Variable	Findings	Level of significance
Logistics organisation (Appendix 1 – Question 3)	City dimension	Cities with fewer inhabitants feel less confident about their capacity to organise urban logistics.	$p < 0.05$
	Position	Mayors and vice-mayors are confident about their capacity to move the necessary means. Assistants/advisors and technicians are more cautious.	$p < 0.01$
Citizens' Data (Appendix 1 – Question 6)	City dimension	People from the cities with less than 25,000 inhabitants were the only ones to answer that they have detailed information about all citizens. Bigger cities indicated they only have on a few.	$p < 0.01$
Cities' Technological Tools (Appendix 1 – Question 7)	City dimension	Most of the respondents of the cities with more than 200,000 mentioned they have tools that allow real-time visualisation and study of scenarios. Smaller cities have responded the opposite.	$p < 0.05$
Real-time Logistics Challenges (Appendix 1 – Question 10)	Position	Mayors and vice-mayors are cautious about their capacity to meet citizens' needs in 15 min. Councillors are more confident.	$p < 0.05$

When asked if cities have citizens' data, such as medical and dietary needs, only 1.95% answered positively. In comparison, 76.26% said they do not have it or only on some citizens. The policy-makers of the most populated cities mentioned they have only data about some citizens, which can be explained by the challenge of gathering detailed data. Another reason could be the fact they might be meticulous because of General Data Protection Regulation (GDPR) issues. Conversely, policymakers from cities with less than 25,000 inhabitants were the ones to point out they have detailed information about all citizens. The close relationship with the residents can explain this. However, there are issues with data structure and digital availability. Thus, their answer can be biased by their lack of knowledge and comprehension about data standardisation as the basis for software systems and applications.

4.1.1. Capacity to meet citizens' needs in real time

When contextualised on the scenario of having to meet the needs of citizens within 15 min (Appendix – question 10), only 2.37% of the respondents said they would not struggle to move the necessary means.

In addition, the challenges raised were mainly due to the lack of logistical planning and coordination and the inexistence of dedicated resources. Table 4 summarises the qualitative analysis of the challenges raised by policymakers to perform a 15-minute response for the last mile.

When asked how cities would depend on private entities (Appendix – question 5), the responses reveal that 51.26% refer to human resources and 27.14% mention the ability to coordinate logistics. In addition, only 11.56% of the respondents referred to delivery vehicles and 10.05% storage spaces (the remaining response options). Thus, standardising tools can support decision-making and fill the existing gap.

4.1.2. Foundation axes of a solution to answer the research questions

The answers to the questionnaire showed that although policymakers feel confident about their capacity to organise the last mile, they do not have technological tools or citizens' data to help them act proactively. Therefore, in addition to the challenges pointed out by policymakers, seven experts were brought together in a focus group to discuss how policymakers can address citizens' needs while reducing the number of vehicles needed in the last mile for logistics activities. The main conclusions drawn from the focus group are highlighted under the features described in Table 5.

Several examples were mentioned during the focus group. First, the delivery platforms that use subcontractors to perform the deliveries were mentioned. These platforms adopted proximity models to restrict consumers from choosing their meals from nearby restaurants for shorter-distance services and faster response.

In addition, software platforms (e.g. OpenStreetMaps) that have detailed information about the territory were also highlighted. Mobility platforms are also focused on providing a quick response from the closest drivers to the clients' location, trying to position them according to historical and real-time information.

There was a general agreement that proprietary supply chains are not prepared to respond to personalised requests in a few minutes. Moreover, to enhance the collaboration between stakeholders and resource sharing, the decentralisation of processes (on the last mile) should be complemented with a centralised logic (virtual) of an integrated open data system. Thus, as represented in Figure 2, instead of the (traditional) supply chain answering reactively after a purchase (where the transport is performed from a distant warehouse, taking several days to fulfil the request, incompatible with instant deliveries), the objective is to focus on the storage of goods in the last mile for a prompt response. Thus, at the purchase moment, the required goods are already stored in a location close to the consumer, allowing smooth modes of transportation to perform the delivery. To achieve this, it is necessary to have detailed data about the citizen profile, preferences, and history for matters of prediction.

Table 4. Challenges for 15 min last-mile fulfilment.

Challenge	Description
Data	• Lack of integration and sharing of data between cities and governmental bodies. • Lack of updated contact databases. • Lack of rigorous and detailed knowledge of all existing resources. • Lack of data collection and computerisation. • Lack of data on possible logistical needs.
Human resources	• An insufficient amount of human resources. • Lack of specialised human resources. • Lack of knowledge of logistics and operations management. • Lack of training for using digital tools to organise and optimise logistics processes. • Lack of autonomy of departments.
Organisation	• Lack of internal organisation, previously defined plans, and established hierarchical organisation for this matter. • Lack of coordination with other local and national entities and private entities. • Difficulty in communicating and coordinating tasks and responsibilities. • Difficulties in the reorganisation of public services because of the rigidity of schedules.
Process	• The high bureaucratic burden of public procurement is unfavourable to emergencies. • Difficult to establish supply contracts (physical acquisition of the necessary goods) and operationalise intermediary logistics. • Struggle to ensure the compliance of the commitment of the different entities involved.
Territory	• Population number and dispersion. • Topography of the territory. • Accessibility issues, namely rural and mountainous areas.
Means	• Lack of delivery means to respond in 15 min. • Low dispersion of storage warehouses. • Struggle to deliver goods that need cold storage.
Tools	• Lack of IT tools to plan and support decision-making. • Difficulty in contacting and communicating with the entire population. • Difficulty in managing information.

Table 5. Features for the development of a solution based on the findings of the focus group.

Feature	Description
Collaborative logistics	The interoperability between public and private entities to promote an omnichannel where all resources are at cities' disposal
Micro logistics operators	Subcontractors or volunteers to provide delivery and storage resources to enhance the capacity of the operation
Flexible and multi-modal modes of transportation	Integration and consideration of multiple types and transportation, positioning them according to the needs
Stock storage within the last mile	Shared storage assets to allow the needed stock to be available in the last mile
Geography of the territory	Software tools that enable the recognition of the territory's topography and accessibility
Forecasting of citizens' needs	Through the forecast, cities can proactively allocate stock to storage points close to citizens in advance to meet their needs within a short period

The intermediate delivery at off-peak hours to the last-mile storage points would proactively ensure the availability of goods in real time. Thus, the number of vehicles in circulation would be reduced. Since there is no time constraint on this supply chain stage, the problem can be

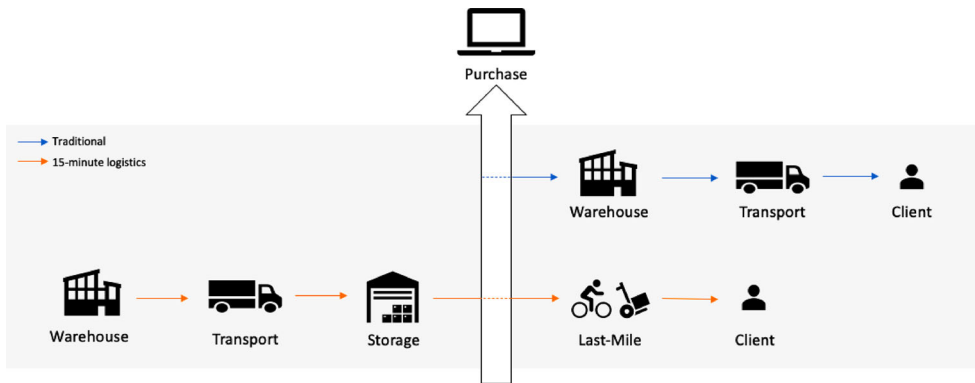


Figure 2. Comparison between the traditional logistic model and the 15-minute last-mile logistical model.

summarised as optimising the maximum travelled distance by the minimum resources. In addition, the final accumulation of last-mile deliveries may be redundant in this model since they will be performed by non-polluting means of transportation (e.g. bicycles, cargo bikes and scooters).

4.2. Proposed model

The literature review highlighted the evolution towards local realisation and real-time urban adjustment based on the study of the most appropriate locations to store goods (on the basis of the spatial indexing of the territory). Thus, it noted that last-mile fulfilment might make use of mobile stock points and transport co-modality, depending on the collaboration schemes among stakeholders and the objectives defined. Furthermore, routing will depend on the choice of the vehicles to perform the deliveries, which should be environmentally friendly. Since the intention is to decrease the number of vehicles in circulation in city centres using smooth modes of transportation to perform the deliveries, is necessary to predict and correctly position the stock towards future demand.

On top of this, the results of the questionnaire acknowledged the inexistence of data collection from municipalities as well as of on-site resources and visualisation tools to provide a response to citizens' needs. Thus, it was notable from the policymakers' responses that there is the need for a tool to help coordinate and support decision-making that takes into consideration the characteristics of the territory and the attributes of existing storage and delivery resources.

In addition some assumptions may also be added from the focus group findings. The proposed model should be based on a collaborative logistics framework to create an omnichannel that uses public and private resources. Thus, the model should consider multiple transportation types and storage facilities to allow storage and delivery of goods within the last mile. The organisation of the stock and the choice of the best-integrated solution should be grounded on the forecast of citizens needs and the topography of the territory. This way, existing stock can be distributed by storage points using shared resources to optimise processes and minimise the associated costs. [Figure 3](#) sketches out the high-level scheme of the proposed model.

4.2.1. Step-by-step description

The designed algorithm considers the following standard steps, represented in [Figure 4](#):

- A. Analysis of historical data – parameterisation of the citizens' locations (and potentially their individualised data).

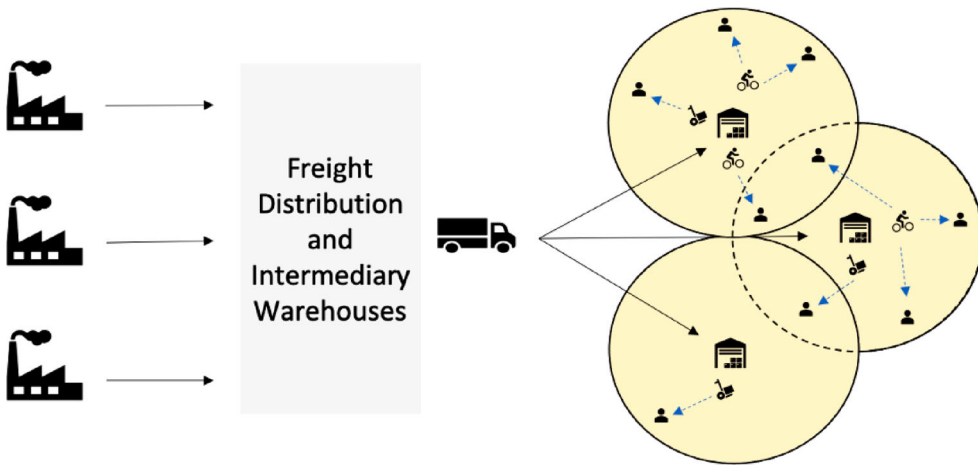


Figure 3. High-level scheme of the proposed model.

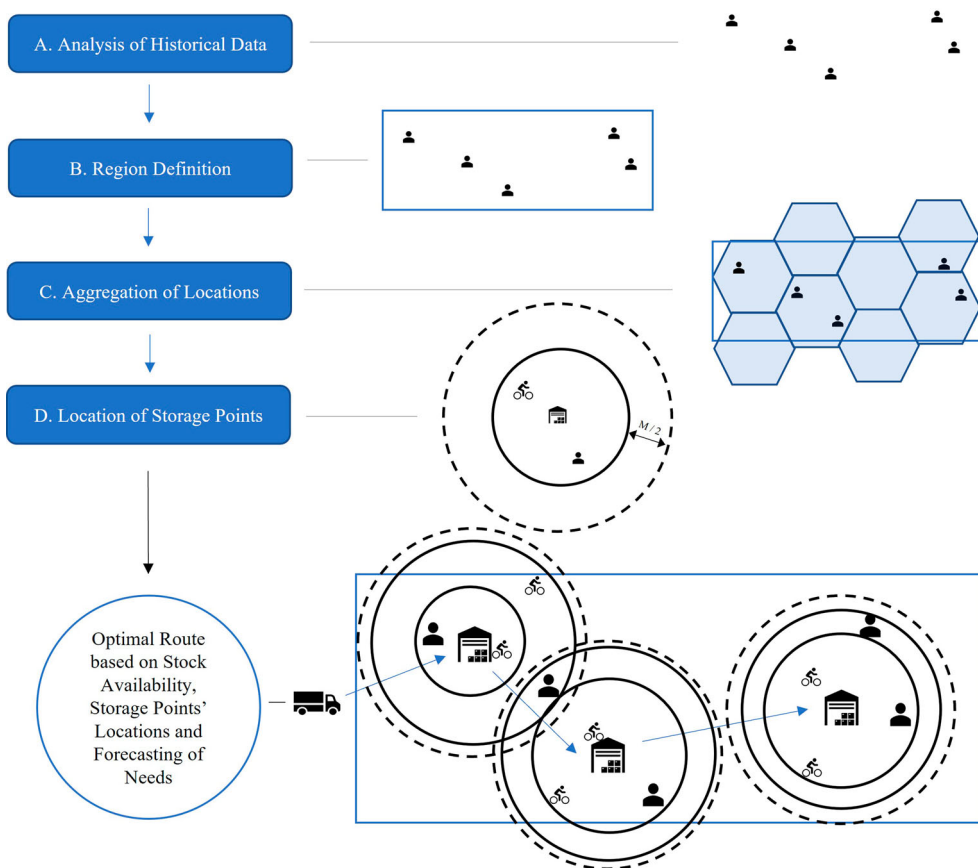


Figure 4. Steps of the designed solution.

- B. Region definition – clustering of locations using a spatial clustering algorithm, and determination of the centroids of each cluster.
- C. Aggregation of locations – spatial indexing for selection of candidate storage points.
- D. Location of storage points – calculation of the maximum attainable region (isochrone).

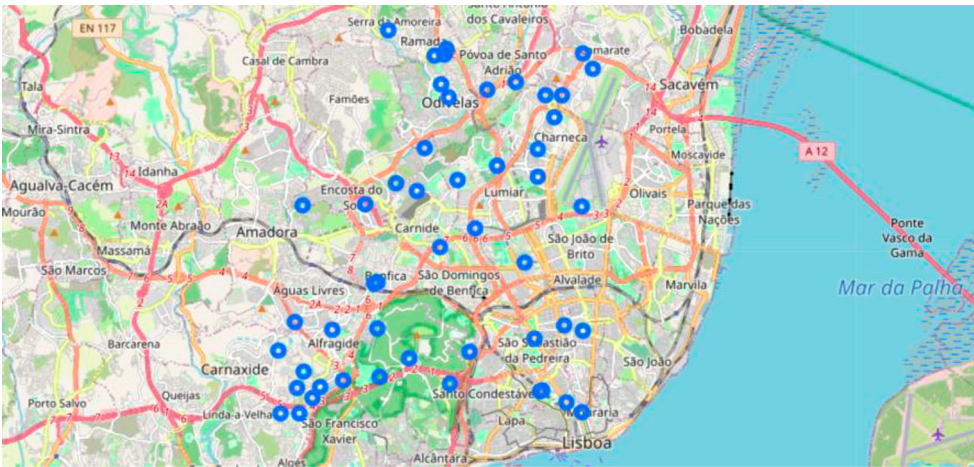


Figure 5. Citizens' locations.

For a generic demonstration of the development of the model, a set of random data was generated. The city of Lisbon, Portugal was used as a test case. This model assumes the cooperation of stakeholders and their resources. Thus, the goal is to position them according to the demand forecast and develop a rationale to choose the best solution to answer in real time.

No specific need or product was not considered. However, it should be recalled that appropriate filtering may be applied, considering the type of goods to be stored and their characteristics. They can be decisive for the optimal result.

a) Analysis of historical data

The initial step is to get the necessary data depending on the use case and its parameters. Therefore, mechanisms to automatise the collection and harmonisation of data of public and private bodies must be considered to ground the model.

As the system should be based on data analysis and forecasting, stock management depends on the expected needs to ensure the necessary goods at a future moment. Moreover, as illustrated in [Figure 5](#), the citizens' locations are parametrised through data analysis. This way, it is possible to make future predictions through machine learning algorithms, e.g. random forest and linear regression (Khaledian and Miller 2020; Poon et al. 2020; Tamiminia et al. 2020). These algorithms should allow the adjustment of the warehouse location in real time (proactive model). The premise will follow a relationship with the existence of local stock. Therefore, calculation of stock should consider the traditional model of Disruption with Instant Replacement – just-in-time (Shah and Ward 2003).

As citizens' specific needs, other attributes can be added to base the model according to the intended use case. For example, in the case that the intention was to provide citizens with their medicines, the system would automatically get each citizen's medical prescriptions. In this case, as it intends to provide a generic scenario, citizens' locations will be considered. This can also be dynamic if the system considers the real-time location of each citizen (e.g. from their smartphone's GPS).

b) Definition of the region

After having the locations (potentially attached to personalised characteristics), their aggregation will allow storage candidate points to be obtained. The first candidate points will be the central

points of the densest regions (clusters). This method is named clustering (and it can be performed considering multiple variables). Moreover, it is done using a clustering algorithm to automatically define regions by aggregating the most significant number of points (Wang et al. 2018). This decision will be dependent on the total number of requests to meet. In this case, 100% of them will be considered, i.e. every citizen's needs must be met.

Thus, the DBSCAN clustering algorithm (Chen et al. 2021) was used since it is a density-based algorithm that does not require the number of clusters in advance. Instead, it selects each point for a cluster according to the point density. In this case, the parameterizable maximum distance of 1.5 km (also known as the epsilon parameter) was chosen to determine whether a location should be included in the cluster. This considers the fact that the initial position of the delivery asset can be a distance of 1.5 km from the storage facility and therefore has to cover twice that distance (in the worst-case scenario). The distance between points is calculated by Haversine's formula (Boeing 2018, 2019). Figure 6 showcases the defined regions.

c) Aggregation of locations

The consideration of clusters (in this case there are five) is vital to reducing the potential locations to analyse. Consequently, the computational effort allows faster analysis (otherwise, the model would have to consider every geographical position in the region).

Nevertheless, the central points of the clusters would be insufficient to find an efficient solution, and therefore more points are needed to run the algorithm. Therefore, additional points within the clusters must be considered. Thus, the aggregation of the initial locations through spatial indexing (within the clusters) allows the final list of candidate storage points to be obtained.

The H3 library (Uber 2021), which divides the Earth's surface into hexagons (and pentagons located in the middle of the ocean), can be used for this task. Furthermore, each H3 index area depends on the desired resolution, ranging from 0.9 square metres to more than 4,250 million square kilometres (Uber 2021). Moreover, resolution 8 was chosen, which corresponds to an area of approximately 737 square metres, representing a more outstanding balance between the number of indexes found and the area of aggregating locations. Next, the candidate point of each index is chosen as its central point, as shown in Figure 7.



Figure 6. Clustering of locations to define existing regions using DBSCAN.

d) Location of storage points

This step relates the storage locations to the delivery. The results are not affected by the delivery time in the previous steps. Thus, this step moves the model from 2D to 3D. It is the most demanding calculation on the computational resource level since it uses geography and topography (as other variables such as traffic) to analyse the distances on the map. This was why previous steps tried to optimise the number of candidate points.

Therefore, after obtaining a set of candidate points (the central points of each cluster obtained via DBSCAN, and the central points of each H3 index, a hexagon, through the aggregation of locations), a chosen mean of transport should be considered to calculate the maximum region attainable in real time (e.g. 15 min). This region is called an isochrone. It was calculated using an open-source route optimisation engine called Valhalla (Belikov and Afonichkina 2021), which uses OpenStreetMaps data to provide the best route between two or more locations, among others. Thus, the calculation is not affected by existing traffic, only by the geography and road information.

The maximum time for the delivery should be defined. In practical terms, half of this value should be used since the border of the isochrone should correspond to the distance it takes to travel half of the maximum time. This covers the worst-case scenario. Thus, it is guaranteed that the temporal restriction is not exceeded. Another essential characteristic is selecting the means of transport to be used, as it will significantly influence the size of each isochrone. In this case, for each candidate point, the respective isochrones were calculated for 15 min, using the bicycle as a means of transport, as illustrated in Figure 8. The result is a polygon that represents the maximum attainable region. Thus, a solution for the location of the storage points is known.

The result shown in Figure 8 indicates the chosen points and their attainable regions within the defined time interval. The choice of each point is made according to the highest coverage rate, and there is the possibility that several regions may intersect, which is the case.

Isochrones can also be refined to find the optimal centroid points (depending on the use case). Different optimisation rules can be considered. Since the characteristics of the needs will be flexible and new historical information will be added to the data set, the storage locations will change based on the system update. Moreover, the locations are constantly corrected based on the forecast of

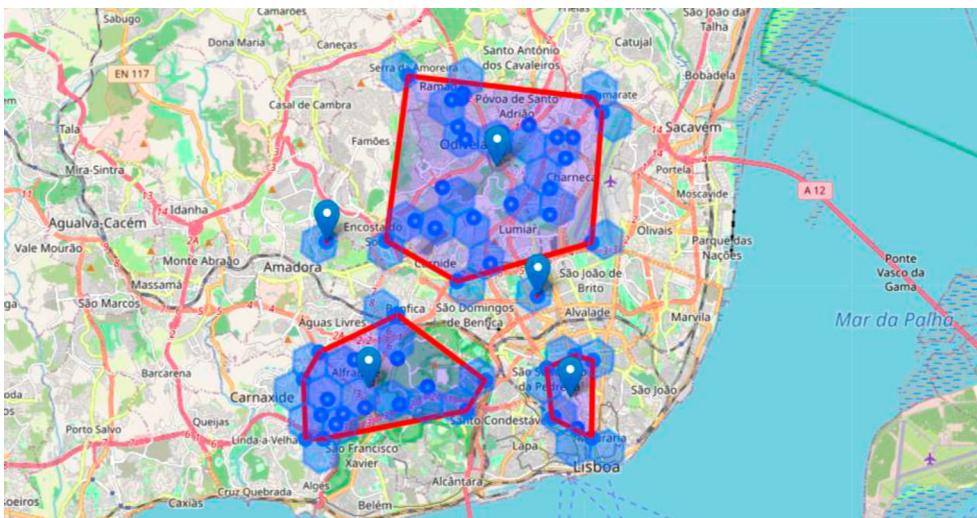


Figure 7. Spatial indexing using H3.

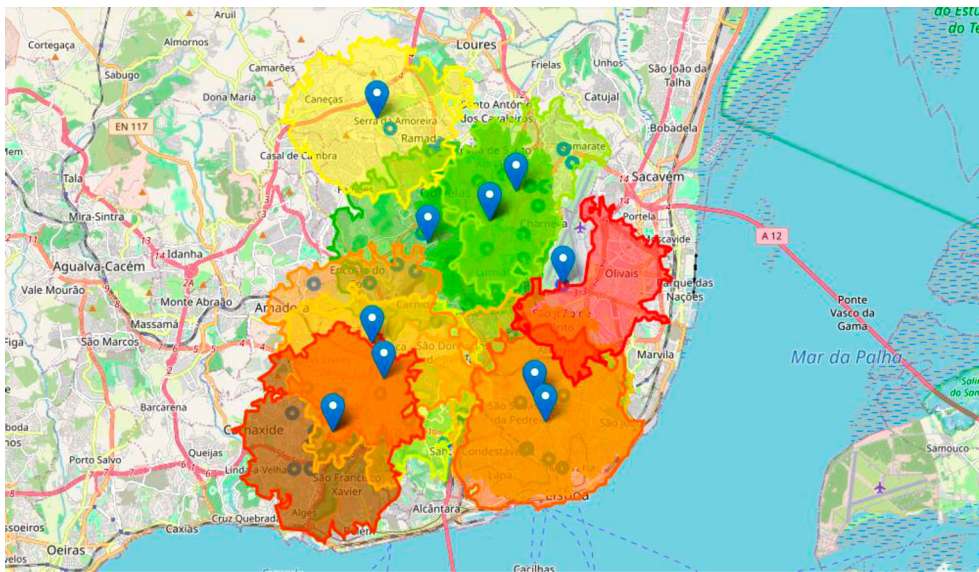


Figure 8. Calculation of the isochrones with Valhalla and OpenStreetMaps.

future needs. These storage points can also be reduced if their mobility meets the system dynamics (in the case of considering innovative vehicles and mobile storage points).

4.2.2. Enhancing the proposed model to a generic perspective

Given the flexibility needed today, beyond studying and realising the fixed location of a logistics centre, it is necessary always to know the best option to satisfy citizens' real-time needs by defining the most appropriate collection pick-up point and respective position.

Moreover, citizens' data must first be structured and analysed to adapt the model to any product or need. Furthermore, defining the restrictions related to a particular use case will help obtain a more reliable solution. Finally, the optimisation rule should be defined according to the objective.

The model proposed was developed and tested on Google Colaboratory (Google 2021a). This algorithm allows for the development and execution of code snippets using the Python programming language, is free to use for research purposes, and is useful for data processing and analysis.

Furthermore, after defining the goal function, whatever the intended use case is, the execution of the following generic standard steps detailed in Figure 9 must be considered:

The application of the model described will naturally decrease the number of polluting means and the kilometres travelled. The placement of storage locations closer to citizens and their real-time adjustment, as represented in Figure 10, will lower the number of visiting points.

5. Discussion

The questionnaire results are in line the findings of Dias et al. (2018), who claimed that cities are not prepared to manage urban freight, as public managers do not possess the knowledge and expertise about urban logistics. The results demonstrated that policymakers do not lack delivery vehicles or storage spaces. Instead, they would depend mostly on private entities because they lack human resources and the ability to coordinate logistics. Nevertheless, integrated urban distribution of goods is critical for municipalities (Oliveira et al. 2018). The capillarity needed for a fast answer demands a greater number of resources, which are not in municipalities' hands.

Inputs:
- Resolution $R = [0.15]$, to be used to retrieve H3 index. - Cluster threshold (CT) in kilometres, maximum distance to include points in a cluster. - Transportation type (T) to be used on isochrone computation – bicycle, car, truck, foot - Maximum travel time (M) to consider
Output: list of points to consider as logistic stock points.
Description:
1 - Initialize location candidates and isochrones list 2 - Get the locations dataset to be analysed 3 - For each location in the dataset, do: 3.1 – H3 index computation with resolution R 3.2 – Store H3 index centroid in the location candidates list 4 – Initialize DBSCAN algorithm parameters: 4.1 – Epsilon parameter = $CT / 6371.0088$ (kilometres by radian) 4.2 – Set distance parameter to Haversine’s algorithm 5 – Compute DBSCAN for the location’s dataset 6 – Store the clusters found centroids in the location candidates list 7 – For each location candidate, do: 7.1 – Compute isochrone for T transport type and M/2 maximum travel time 7.2 – Store the isochrone computed on the isochrones list ... 8 - Choose the best points

Figure 9. Algorithm commands.

Furthermore, the results suggest that although policymakers feel confident in their capacity to organise the last mile, they would struggle to operate autonomously. Therefore, it is evident they need the dynamic cooperation of other stakeholders. In addition, the results demonstrated that there is still a lack of tools that enable real-time visualisation and study of scenarios for placing

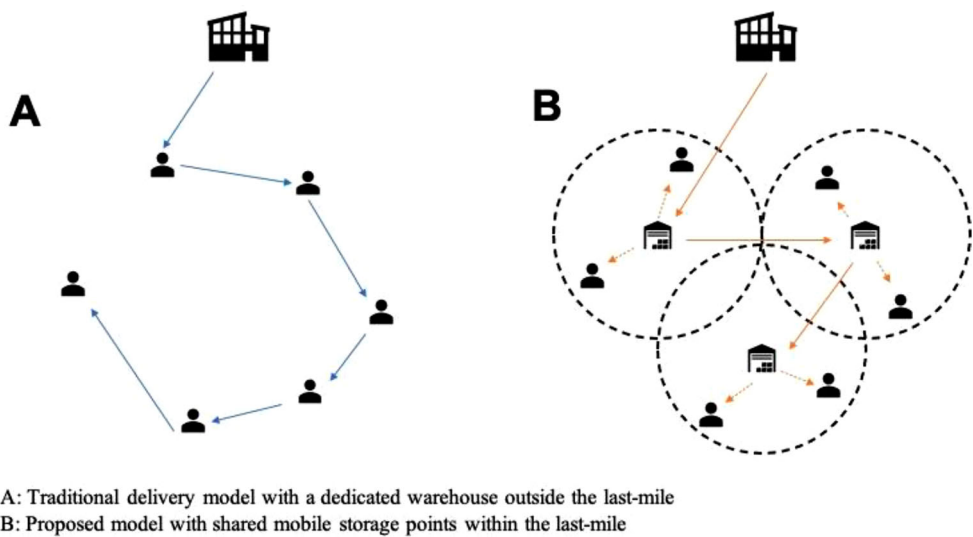


Figure 10. Traditional logistics vs proposed model.

and organising delivery and storage means. This discussion may be enhanced by reflecting on who should organise the last mile.

As stated in the literature, data-driven logistics models depend on collaboration schemes among stakeholders and the defined objectives (Dolati Neghabadi, Espinouse, and Lionet 2021; Gläser, Jahnke, and Strassheim 2021; Sundarakani, Ajaykumar, and Gunasekaran 2021). Moreover, if the objective is to fulfil a common good and a societal need, cities may have the ability to coordinate an integrated answer. Nevertheless, even if the subject is the usual economic activities, experience shows that it is hard for private entities to adopt an open data mindset and share business insights with competitors; moreover it may not be viable to let them and other stakeholders decide on how to manage existing resources to provide a real-time answer, since it may continue to challenge cities' sustainability. In fact, Correia, Teixeira, and Marques (2021c) noted the lack of capacity to perform instant deliveries and personalised requests in a case study with 74 Portuguese e-commerce companies. Thus, it is crucial to review and discuss cities' role in standardising and integrating data to optimise last-mile urban logistics while decreasing the number of existing resources. This discussion must be enhanced by the urgent necessities of combating greenhouse gas emissions and meeting citizens' needs.

Based on the questionnaire results, the challenges posed by policymakers to respond in 15 min can be summarised in the following:

1. Lack of integration/sharing of updated data between stakeholders;
2. Lack of specialised human resources with knowledge of logistics and operations management;
3. Lack of coordination between departments and responsibilities among public bodies;
4. Lack of commitment from private sector and efficient public procurement;
5. Dispersion and topography of the territory;
6. Lack of delivery and storage means;
7. Lack of tools to centralise and communicate information.

The insights from the focus group are aligned with the trend of considering a sharing economy open cross-company networks, based on win-win, last-minute, business collaboration schemes to optimise urban transport, using real-time data to finding the best locations for storage while using smooth moods of transportation for the delivery (Ducret 2014; Kirch, Poenicke, and Richter 2017; Nathanail, Gogas, and Adamos 2016; Oztemel and Gursev 2020; Rudolph et al. 2022). However, it does not fully agree with Ghaderi et al. (2022) who consider people's daily travels instead of dedicated deliveries. Their model would require greater computational and logistics efforts to obtain the best options available. Citizens can perform the deliveries but have to input their exact availability and resources without rigidly restricting the model to their planned trips. Operationally, it would be challenging to provide a personalised request, since they would not have much time-window flexibility for delays and unexpected events.

Furthermore, the proposed algorithm of this research is aligned with the future breakdown of the supply chain to the last mile. Thus it is developed under the umbrella of the Logistics 4.0 concept, where the sharing economy has a vital role and seeks the best logistics instantaneous relationship option within existing marketplaces towards the fulfilment of personalised requests (Correia, Teixeira, and Marques 2021a; Daugherty, Bolumole, and Grawe 2019; Dolati Neghabadi, Espinouse, and Lionet 2021; Gläser, Jahnke, and Strassheim 2021; Lim, Jin, and Srai 2018; Ranieri et al. 2018; Rimmer and Kam 2018; Shi, Liu, and Zhang 2021; Srivatsa Srinivas and Marathe 2021; Sundarakani, Ajaykumar, and Gunasekaran 2021; Taniguchi, Thompson, and Yamada 2016). This model is inspired by the creation of a resource sharing logistics emergency network as proposed by Wang, Peng, and Xu (2021). Nevertheless, instead of considering local multi-centre storage with time-windows as proposed by Wang et al. (2022), the model's first assumption is the input of maximum specific time for delivery which must define the allocation of resources inside the respective

marketplace., something that already happens in other use cases as on the luggage sector by the start-up LUGGit (2022).

Thus, to enable instant deliveries and therefore meeting real-time citizens' needs, it is crucial on one side to understand the specifications of their demand and, on the other side to find the best solution available to fulfil personalised requests based on their characteristics. Since there is stock to organise and allocate, it is vital to characterise the population's needs and apply forecasting algorithms to position storage assets according to future needs.

Furthermore, to achieve a real-time answer (within a period of 15 min) storage decentralisation is vital. Thus, considering every storage space and delivery means available will ground dynamic logistics schemes and reduce distances travelled. The storage points can be warehouses, garages, or other facilities (public or private). Therefore, the developed algorithm assumes greater importance since it presents the first step on which to base a dynamic and collaborative system. With the optimal locations of storage points, policymakers can organise the stock (of essentials in the first place) and promote local business relationships and products. Thus, with the cooperation of stakeholders, it would be possible to decrease their costs on operations and logistics needs, which will also impact the product's final cost. This inclusive and local approach will help the microeconomic aspect and respond to the macroeconomic context of inflation. Additionally, it can also promote the inclusion of citizens by responding to the ageing and isolation of the population because of the study of the locations and local resources to fulfil their needs.

Although cities have been working to gather data on citizens and city infrastructure, logistics data is scarce. Categorisation of logistics assets such as vehicles and warehouses is critical. Collection, transmission, and storage of data is also vital. The sharing of resources would pose challenges that need the creation of standards to integrate data, hardware, and communications. Moreover, stakeholders' data should be open so municipalities can always know existing resources, locations, and capacities. This way it would be possible to set up logistics schemes to guarantee that a real-time response could be provided to any necessity in a pre-defined period. Thus, in-depth discussion is also needed on these technologies. A common city application could be provided, where entities could resort to available logistics assets. Software tools must be created and integrated into existing smart city urban platforms.

Each company or individual should automatically update the system with their current resources and tasks for public matters. A minimum threshold per entity to meet citizens' needs can be defined. The line begins to be tenuous between the business models associated with industry and the role of cities and policymakers. Nowadays, it is impossible to reflect on industry without thinking about city planning and vice-versa, since one directly impacts the other. This fact should promote the reflection on policymakers' role as regulators and integrators and the measures to be taken to allow data integration.

Furthermore, if the goal is to promote cities' sustainability by the cooperation and integration of stakeholders, and if this depends on the interoperability between systems, there is a need to encourage stakeholders to adopt standards and share their resources data in real time. Thus, policymakers need to guarantee the privacy of existing economic activities to allow policymakers in a later stage to organise their assets based on existing demand. This strategy can start with an early-adopter sector and a specific citizen need. Thus, privacy and cybersecurity are some of the most significant challenges for future developments. Economic activities should not be harmed by stakeholders' transparency towards improving urban logistics and cities' sustainability.

6. Conclusion and future work

Cities are being held hostage to the evolvment of e-commerce and the spread of entities vehicles in urban areas. The tendency towards personalisation of industry will continue to put pressure on the last mile and concentrate delivery vehicles and warehouses. Thus, proprietary supply chains will continue to challenge cities' sustainability if policymakers do not proactively manage the

integration of resources. Smart cities emerged to provide policymakers with a common operating picture to support their decisions. Nevertheless, this is still scarce for the organisation of last-mile urban logistics. Therefore, cities must take a greater interest in promoting the entities' integration and in developing of collaborative tools since this will significantly impact the existing quality of life and sustainability.

This study showed that policymakers recognise the importance of having a tool that enables real-time simulation to place provisional means (delivery and storage) for the supply of goods. Furthermore, considering the mentioned barriers about the lack of human resources in the structure of cities' and the capacity for organisation pointed out by policymakers, an initial algorithm was designed to support decision-making. It considered collaborative logistics, flexible and multi-modal modes of transportation, stock storage within the last mile, the geography of the territory, and the forecasting of citizens' needs.

The proposed four-step logistics algorithm, on the one hand, allows the improvement of urban logistics by considering the cooperation of stakeholders, thus decreasing the amount of resources and labour needed, and on the other hand, allows a real-time personalised response to citizens' needs based on the positioning adjustment and choice of the resources to perform the deliveries.

Nevertheless, policymakers raised several challenges about cities' capacity to meet citizens' needs within 15 min regarding data, human resources, organisation, process, territory, means, and tools. Thus, the challenges pointed out supply some of the takeaways of this research, which are: (1) the need to educate the population and policymakers about the importance of data integration to work together with the economic agents to find ways to address current challenges; (2) methods and standards to collect citizens' data; and (3) that regulatory frameworks must be created to guide the cooperation of economic agents.

The case study carried out within this research marks an initial perspective and brings valuable insights for broader discussion in terms of the role of cities in organising last-mile urban logistics and the need for tools to promote the collaboration of stakeholders. The methodology applied highlighted the existing gap throughout the Portuguese territory regarding data accessibility and cities' response capacity. Furthermore, this study can be applied in other countries where the proposed model will follow the same structure to operationalise an integrated answer to citizens' needs.

As future work, the focus on the last-mile collaboration and sharing of resources between stakeholders can be enhanced to the entire supply chain. Thus, the first step would be to allow different companies of the same sector to standardise the way they consider their resources and their communication. This way, they could put their vehicles and warehouses at the disposal of the common good to increase the capillarity and efficiency of their resources. Furthermore, it will be crucial to study how cities gather citizens' data and treat logistics information about existing resources to standardise technical frameworks. After that, having a unique visualisation software will allow others to join. At the same time, policymakers can access this and establish service provider contracts to operate with the combination of proactively available means. This will significantly impact the vision of cities' operations and capacity to meet citizens' needs.

In terms of limitations, this study does not consider that different policymakers from the same city may have different perspectives. In addition, the proposed algorithm assumed the cooperation of stakeholders and sharing of their resources. Thus, it did not focus on the technical constraints of integrating the stakeholders' solutions into city infrastructure. Future work needs to emerge about implementation of the steps.

Data availability

The authors confirm that the data supporting the findings of this study are available within the article and its supplementary materials.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendix

Table A1. Questionnaire.

Group	Question	Answers
I	Identification of the respondent	
	(1) What is your gender?	• Female • Male
	(2) How old are you?	• 18–25 years old • 26–39 years old • 40–49 years old • 50–64 years old • + 65 years old
II	Scenario 'The emergence of a new pandemic'	
	(3) In your opinion, do you think that cities would be able to organise last-mile urban logistics to meet citizens' needs?	• 1 – No • 2 – Not much • 3 – More or less • 4 – Yes, totally
	(4) How dependent would you be on the responsiveness of the private sector?	(Likert Scale) • 1 – Not at all dependent • 5 – Very dependent
	(5) If you did not respond with the value 1 – 'Not at all dependent', the existing dependency would be mainly at the level of:	• Delivery vehicles • Storage warehouses • Support Human Resources • Capacity to coordinate logistics • Other option (Open)
	(6) Do cities have detailed information on citizens and their individual needs (example: address, health care, medical prescriptions, and food concerns)?	• No • Of some • Of the majority • Yes, of all
	(7) Do municipalities have technological tools that allow real-time visualisation and the study of scenarios to place provisional means (delivery and storage) for the supply of goods?	• Yes • No
	(8) In case the answer is 'No', how important do you think the existence of a tool with these features would be?	(Likert Scale) • 1 – Not at all important • 5 – Very important
	(9) If you wanted to find the best location to place storage spaces for essential goods to meet a specific number of citizens, how fast could you do it accurately?	(Likert Scale) • 1 – Not at all fast • 5 – Very fast
	(10) Considering that citizens' needs had to be met in real-time (15 min), how difficult would it be for the city to move the necessary means?	(Likert Scale) • 1 – Not at all difficult • 5 – Very difficult
	(11) What would be the greatest difficulties? Explain briefly	(Open Question)

(Continued)

Table A1. Continued.

Group	Question	Answers
III Current Urban Planning	(12) Do you belong to the structure of a city hall?	• Yes • No
	(13) If 'Yes', what is your current position?	(Open Question)
	(14) How many inhabitants does the city have?	• Less than 25,000 inhabitants • Between 25,001 and 50,000 inhabitants • Between 50,001 and 100,000 inhabitants • Between 100,001 and 200,000 inhabitants • Between 25,001 and 50,000 inhabitants • More than 200,000 inhabitants